Veronese Descent Rigidity of the Petersen Equiangular Tight Frame

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Abstract

The Petersen association scheme on $\binom{5}{2}$ vertices has a distinguished primitive idempotent E_1 whose normalised version realises, up to orthogonal equivalence and relabelling, the classical 10-line equiangular tight frame (ETF) in $\mathbb{R}^5$ with inner-product modulus $1/3$. We prove that this ETF admits an essentially unique Veronese-image realisation in $\mathrm{Sym}^2_0(\mathbb{R}^3) \cong \mathbb{R}^5$; equivalently, the projective $\mathbb{RP}^2$-configuration whose quadratic lift produces the Petersen ETF is unique up to $O(3) \times S_5$. The proof combines a representation-theoretic descent argument (Schur's lemma together with a spectral obstruction on the Veronese surface) with an exact computer-assisted algebraic certificate for the rigidity of the classical compound of five tetrahedra under a K_5-incidence constraint. As a corollary we obtain the **Signed Gram Rigidity Theorem**: every real symmetric matrix $H \in \mathbb{R}^{10\times 10}$ with diagonal 1, off-diagonal moduli in $\{\sqrt{5}/3,\ 1/3\}$ distributed by the Petersen pattern, positive semidefinite and of rank 3, equals the icosahedral antipodal face-normal Gram matrix up to switching $D \in \{\pm 1\}^{10}$ and Petersen-graph automorphism $\sigma \in \mathrm{Aut}(K(5,2)) = S_5$.

Keywords: Petersen graph, equiangular tight frame, Veronese embedding, association scheme, regular two-graph, icosahedral symmetry, Signed Gram rigidity.

MSC 2020: 05E30 (association schemes), 05C50 (graphs and linear algebra), 52C35 (arrangements of points, flats, hyperplanes), 42C15 (general harmonic expansions, frames), 15B48 (positive matrices and their generalizations), 20C15 (ordinary representations and characters).

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1. Introduction

1.1 Setting

The icosahedron determines, via its 10 pairs of antipodal face-normals, a configuration of 10 projective points in $\mathbb{RP}^2$ whose pairwise inner products take only two values: $|v_i \cdot v_j| \in \{1/3,\ \sqrt{5}/3\}$. The graph defined by the larger value ($i \sim j \iff (v_i \cdot v_j)^2 = 5/9$) is the **Petersen graph** $K(5,2)$. Squaring the inner products and centring gives a 10×10 matrix K that we identify with twice the primitive idempotent E_1 of the Petersen association scheme; its normalised square root is the classical maximal 10-line equiangular system in $\mathbb{R}^5$ with common angle arccos(1/3) [11, 8, 10]. General absolute and linear-programming bounds for systems of lines and spherical codes go back to Delsarte–Goethals–Seidel [5, 6].

This paper proves that the chain

$$\underbrace{K(5,2)}_{\text{Petersen}} \longrightarrow \underbrace{E_1}_{\text{idempotent}} \longrightarrow \underbrace{\text{ETF in } \mathbb{R}^5}_{\text{Layer B}} \longrightarrow \underbrace{\nu_2(\mathbb{RP}^2)}_{\text{Veronese descent}} \longrightarrow \underbrace{\text{icosahedral config}}_{\text{rigid}}$$

is rigid all the way down: any rank-3 Veronese realisation of the Petersen ETF is $O(3) \times S_5$-equivalent to the icosahedral face-normal configuration, and the corresponding signed Gram matrix is unique up to switching and Petersen automorphism.

1.2 Statement of the main theorems

Theorem 1 (Petersen ETF as primitive-idempotent realisation). Let A be the Petersen adjacency matrix and J the all-ones matrix. The squared-Gram matrix

$$Q = \tfrac{8}{9}\, I + \tfrac{4}{9}\, A + \tfrac{1}{9}\, J$$

is forced by the Petersen-magnitude conditions, and the centred Veronese-Gram

$$K = \tfrac{3}{2}\left(Q - \tfrac{1}{3}J\right) = \tfrac{4}{3}I + \tfrac{2}{3}A - \tfrac{1}{3}J$$

equals $2\,E_1$, where E_1 is the primitive idempotent of the Petersen association scheme corresponding to the eigenvalue 1 of A.

Theorem 2 (Veronese descent rigidity). Let $v_0, v_1, \ldots, v_9 \in S^2 \subset \mathbb{R}^3$ be unit vectors satisfying the Petersen magnitude pattern,

$$\begin{aligned} |v_i \cdot v_j| &= \sqrt{5}/3 \quad \text{on Petersen edges,} \\ |v_i \cdot v_j| &= 1/3 \quad \text{on complement edges } (i \neq j), \end{aligned}$$

and assume moreover that the vectors span $\mathbb{R}^3$ — equivalently, the Gram matrix VV^T has rank exactly 3, where V is the 10×3 matrix with rows v_i. Set $U_i := \nu_2(v_i) = v_i v_i^\mathsf{T} - \frac{1}{3}I$. The normalised lift $\{\hat{U}_i = \sqrt{3/2}\, U_i\}_{i=0}^{9}$ then realises the 10-line Petersen ETF in $\mathbb{R}^5$ (Theorem 1 and Corollary 2.2), and the projective configuration $\{[v_i]\} \subset \mathbb{RP}^2$ is $(O(3) \times S_5)$-equivalent to the icosahedral face-normal configuration.

Theorem 3 (Signed Gram Rigidity, corollary). Let $H \in \mathbb{R}^{10\times 10}$ be a real symmetric matrix satisfying (i) $H_{ii} = 1$; (ii) $|H_{ij}| = \sqrt{5}/3$ on Petersen edges; (iii) $|H_{ij}| = 1/3$ on complement edges; (iv) $H \succeq 0$; (v) $\mathrm{rank}(H) = 3$. Then there exist $D = \mathrm{diag}(\varepsilon_0, \ldots, \varepsilon_9) \in \{\pm 1\}^{10}$ and $\sigma \in \mathrm{Aut}(K(5,2)) = S_5$ such that

$$H = P_\sigma \, D \, H_{\mathrm{ico}} \, D \, P_\sigma^{\mathsf{T}},$$

where H_{ico} is the icosahedral face-pair Gram matrix.

1.3 The proof cascade

The proof cascade is shown in Figure 1. The two middle arrows together constitute the proof of Theorem 2: the **five-tetrahedra hinge bridge** (Theorem 5.6, §5) installs the A_5-symmetry, and the **equivariant Veronese descent** (Theorem 4.3, §4) uses Schur's lemma and the spectral obstruction to upgrade it to uniqueness up to $O(3)$.

Theorem 1 is elementary algebraic-combinatorial (Bose-Mesner algebra). The ETF uniqueness step from E_1 to the abstract 10-line system in $\mathbb{R}^5$ is classical [8, 10, 2]. The genuinely new content is Theorem 2 — *the Veronese descent rigidity.* Theorem 3 then follows by combining Theorem 2 with the standard Gram factorisation.

1.4 What is new

The Petersen ETF and its uniqueness in $\mathbb{R}^5$ are classical. The Veronese model and the equiangular-line interpretation are well known. What we contribute is:

1. A clean **representation-theoretic descent** for the equivariant case (Section 4): Schur's lemma combined with a spectral obstruction on the Veronese surface in $\mathrm{Sym}^2_0(\mathbb{R}^3)$.
2. A **rigidity proof for the compound of five tetrahedra under K_5-incidence** (Section 5): explicit hinge parameterisation, Labeling Reduction Lemma, and an exact computer-assisted certificate reducing 1024 discrete cases to a single quadratic $4s^2 - \sqrt{3}\,s - 3/4 = 0$ in $\mathbb{Q}(\sqrt{3}, \sqrt{5})$.
3. The **combination of (1) and (2)** to obtain a complete proof of Theorem 2 with no further conjectural hypothesis.

The signed Gram rigidity (Theorem 3) was previously verified by an exhaustive sign-pattern enumeration; the present proof recasts it as a corollary of the Veronese descent rigidity, providing a conceptually cleaner story.

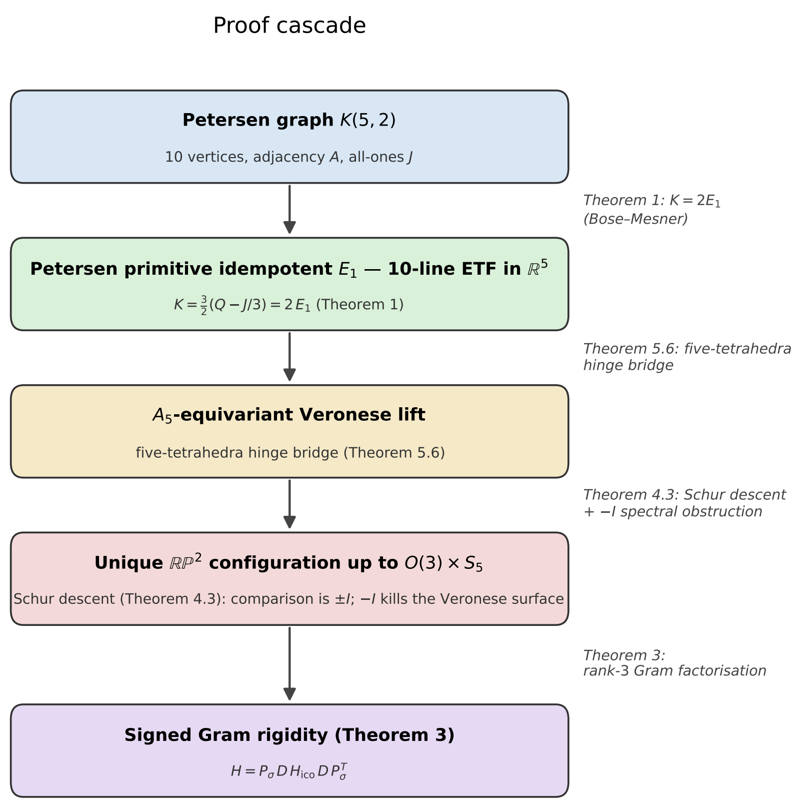

Figure 1: **Figure 1.** The proof cascade. The Petersen primitive idempotent gives the classical 10-line ETF in $\mathbb{R}^5$ (Theorem 1). The five-tetrahedra hinge bridge (Theorem 5.6) installs an A_5-equivariant Veronese lift; Schur's lemma together with the $-I$ spectral obstruction on the Veronese surface (Theorem 4.3) upgrades this to uniqueness up to $O(3) \times S_5$. The Signed Gram Rigidity Theorem (Theorem 3) is the Gram-factorisation corollary.

2. The Petersen scheme and its 5D equiangular tight frame

2.1 The Petersen association scheme

Let the Petersen graph be the Kneser graph $K(5,2)$: vertices are the $\binom{5}{2} = 10$ two-element subsets of $\{1,\dots,5\}$, edges join disjoint subsets. The adjacency matrix A has spectrum $\{3,\, 1^{(5)},\, (-2)^{(4)}\}$. Together with the identity I and the complement adjacency $\bar{A} = J - I - A$, these form the Bose-Mesner algebra of the **Petersen association scheme**, with primitive idempotents

$$E_0 = \tfrac{1}{10}\,J, \qquad E_1\ (\dim 5), \qquad E_2\ (\dim 4),$$

where E_1 projects onto the 1-eigenspace of A and E_2 onto the (-2)-eigenspace.

2.2 The squared-Gram matrix and the centred lift

Suppose 10 unit vectors $v_0,\dots,v_9 \in S^2$ satisfy the Petersen-magnitude pattern: $|v_i \cdot v_j| = \sqrt{5}/3$ on Petersen edges, $|v_i \cdot v_j| = 1/3$ on complement edges. Then the entrywise square

$$Q_{ij} := (v_i \cdot v_j)^2$$

equals 1 on the diagonal, 5/9 on Petersen edges, 1/9 on complement edges. In matrix form:

Proposition 2.1. *$Q = \frac{8}{9}\,I + \frac{4}{9}\,A + \frac{1}{9}\,J$.*

Define the centred lift

$$K := \tfrac{3}{2}\left(Q - \tfrac{1}{3}J\right) = \tfrac{4}{3}\,I + \tfrac{2}{3}\,A - \tfrac{1}{3}\,J.$$

2.3 The primitive-idempotent identification

Theorem 1. *$K = 2\,E_1$.*

Proof. The Petersen Bose-Mesner algebra is spanned by I, A, J; equivalently by E_0, E_1, E_2. Decomposing $\mathbb{R}^{10}$ into the three eigenspaces:

- On the $\mathbf{1}$-line (eigenspace of J): A acts as 3, J as 10. Apply K: $\frac{4}{3}\cdot 1 + \frac{2}{3}\cdot 3 - \frac{1}{3}\cdot 10 = \frac{4}{3} + 2 - \frac{10}{3} = 0$.
- On the A-eigenspace for $\lambda = 1$ (dimension 5, in $\ker J$): A acts as 1, J as 0. Apply K: $\frac{4}{3} + \frac{2}{3}\cdot 1 - 0 = 2$.
- On the A-eigenspace for $\lambda = -2$ (dimension 4, in $\ker J$): A acts as -2, J as 0. Apply K: $\frac{4}{3} + \frac{2}{3}\cdot(-2) = 0$.

So K vanishes outside the 1-eigenspace of A and acts as twice the identity on it: $K = 2\,E_1$. $\square$

2.4 The normalised lift and the ETF structure

Let $U_i := v_i v_i^\mathsf{T} - \frac{1}{3} I \in \mathrm{Sym}_0^2(\mathbb{R}^3) \cong \mathbb{R}^5$ (the trace-free part of the Veronese image). A direct computation gives $\|U_i\|^2 = 2/3$, so the *unit-normalised* lift is $\hat{U}_i := \sqrt{3/2}\, U_i$, and

$$\langle \hat{U}_i, \hat{U}_j \rangle = \begin{cases} 1, & i = j, \\ +1/3, & (i,j) \in E(\text{Petersen}), \\ -1/3, & (i,j) \in E(\overline{\text{Petersen}}),\ i \neq j. \end{cases}$$

Equivalently $K_{ij} = \langle \hat{U}_i, \hat{U}_j \rangle$, confirming that $K = 2E_1$ is precisely the Gram matrix of the normalised Veronese-lift.

> **Corollary 2.2 (Tight-frame property).** The normalised lift $\{\hat{U}_i\} \subset \mathbb{R}^5$ is a **tight frame** with frame bound 2:
>
> $$\sum_{i=0}^{9} \hat{U}_i \hat{U}_i^\mathsf{T} = 2\, I_5.$$
>
> The 10 lines $\mathbb{R}\hat{U}_i$ are equiangular with $|\cos\theta| = 1/3$. By the classical Lemmens–Seidel classification recalled in Theorem 2.3, this is the maximal equiangular-line count in $\mathbb{R}^5$.

Proof. Arrange $\{\hat{U}_i\}$ as the rows of a 10×5 matrix Y. By construction $K = YY^\mathsf{T}$. By Theorem 1, $K = 2E_1$ has rank 5 and nonzero spectrum $\{2^{(5)}\}$. Since YY^T and $Y^\mathsf{T}Y$ share their nonzero eigenvalues, the symmetric matrix $Y^\mathsf{T}Y \in \mathbb{R}^{5\times 5}$ has spectrum $\{2^{(5)}\}$ and hence $Y^\mathsf{T}Y = 2\, I_5$. This is the tight-frame relation. □

For the general theory of (finite) tight frames and their equiangular specialisations, the standard reference is Waldron [12].

2.5 Classical uniqueness of the 10-line ETF

> **Theorem 2.3 (van Lint-Seidel, Lemmens-Seidel, Taylor).** The 10-line equiangular line system in $\mathbb{R}^5$ with common angle $\arccos(1/3)$ is unique up to orthogonal equivalence and relabelling; equivalently, the regular two-graph on 10 vertices with Seidel spectrum $\{3^{(5)}, (-3)^{(5)}\}$ is unique up to switching.

We use this as a black box; the references [11, 8, 10, 9, 2] provide the original proofs and an extended modern treatment. The ($\alpha = 1/3$, $N = 10$, $d = 5$) case is the maximum case of the classification of equiangular line systems with common angle $\arccos(1/3)$ in Lemmens–Seidel [8]; Taylor [10] and Seidel's survey [9] set out the regular two-graph formalism and the order-10 uniqueness; Brouwer–Van Maldeghem [2] gives a modern textbook treatment. The connection between equiangular line systems, regular two-graphs, and root systems is the subject of Cameron–Goethals–Seidel–Shult [3].

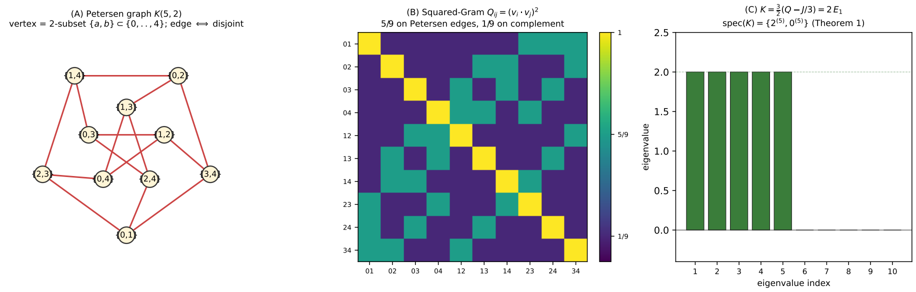

Figure 2: **Figure 2.** The Petersen association scheme layer. **(A)** The Petersen graph $K(5,2)$: vertices are the $\binom{5}{2} = 10$ 2-subsets of $\{0,\ldots,4\}$, edges join disjoint subsets. **(B)** Heatmap of the squared-Gram matrix $Q_{ij} = (v_i \cdot v_j)^2$, taking value $5/9$ on Petersen edges and $1/9$ on the complement. **(C)** Spectrum of the centred lift $K = \frac{3}{2}(Q - J/3) = 2\,E_1$, which is $\{2^{(5)}, 0^{(5)}\}$ — the rank-5 primitive idempotent realisation of the 10-line ETF in $\mathbb{R}^5$ (Theorem 1).

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3. The Veronese model

3.1 Definition

The trace-free Veronese map

$$\nu_2 : \mathbb{R}^3 \to \mathrm{Sym}^2_0(\mathbb{R}^3), \quad \nu_2(v) := vv^\mathsf{T} - \tfrac{1}{3}I$$

satisfies $\nu_2(-v) = \nu_2(v)$, hence factors through $\mathbb{RP}^2$. For unit v, the image $\nu_2(v)$ is trace-free with eigenvalues $(2/3,\ -1/3,\ -1/3)$.

3.2 Spectral characterisation

> **Lemma 3.1 (Veronese spectral characterisation).** A trace-free symmetric matrix $U \in \mathrm{Sym}^2_0(\mathbb{R}^3)$ is in the image of ν_2 if and only if its spectrum is $(2/3,\ -1/3,\ -1/3)$.

Proof. ($\Rightarrow$) Direct from vv^T having spectrum $(1,0,0)$ for unit v.

($\Leftarrow$) If $\mathrm{spec}(U) = (2/3,\ -1/3,\ -1/3)$, then $U + I/3$ has spectrum $(1,0,0)$: rank-1 PSD, so equals vv^T for a unique $\pm v \in S^2$. Therefore $U = \nu_2(v)$. $\square$

This lemma is the **key non-linear constraint** that distinguishes Veronese images from arbitrary trace-free symmetric matrices, and will power the spectral obstruction in Section 4.

3.3 Normalisation conventions

For clarity in the rest of the paper we use:

Object	Notation	Norm	Inner product values
Raw Veronese lift	$U_i = \nu_2(v_i)$	$\|U_i\|^2 = 2/3$	$\langle U_i, U_j\rangle \in \{\pm 2/9\}$
Normalised lift	$\hat{U}_i = \sqrt{3/2}\, U_i$	$\|\hat{U}_i\| = 1$	$\langle \hat{U}_i, \hat{U}_j\rangle \in \{\pm 1/3\}$
Line angle	$\mathbb{R}\hat{U}_i$	–	$\lvert\cos\theta\rvert = 1/3$

Theorem 2 and its proof use both representations; the normalised one makes the ETF/tight-frame language transparent, while the raw one is natural for the Veronese surface.

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4. The equivariant descent theorem

4.1 The A_5-action and the 5D irreducible representation

The icosahedral rotation group is isomorphic to A_5. Choosing one of the two faithful 3-dimensional real irreducible representations $\mathbf{3}, \mathbf{3}'$ of A_5 gives an embedding $A_5 \hookrightarrow SO(3)$; the other is obtained by precomposing with the outer automorphism that interchanges the two 5-cycle conjugacy classes. In either case, applying Sym^2 gives an action on $\mathrm{Sym}^2(\mathbb{R}^3) = \mathbf{1} \oplus \mathbf{5}$, and the trace-free part $\mathrm{Sym}^2_0(\mathbb{R}^3)$ carries the **unique 5-dimensional irreducible real A_5-representation**.

The Petersen graph has abstract automorphism group $\mathrm{Aut}(K(5,2)) = S_5$ (classical; see e.g. [7], [1]). In the icosahedral geometric realisation, the orientation-preserving rotations form the subgroup $A_5 \subset S_5$, acting on the 10 face-pair axes by rotations of $\mathbb{R}^3$. The 10-vertex permutation representation of A_5 decomposes as

$$\rho^{(10)}_{\mathrm{perm}} \;\cong\; \mathbf{1} \oplus \mathbf{4} \oplus \mathbf{5}.$$

In particular, **the 5-dimensional irrep occurs with multiplicity exactly one** in $\rho^{(10)}_{\mathrm{perm}}$.

Verification. The character of $\rho^{(10)}_{\mathrm{perm}}$ takes values $\chi(e) = 10$, $\chi(\text{ord-2 class}) = 2$, $\chi(\text{ord-3 class}) = 1$, $\chi(\text{ord-5 classes}) = 0$. This is the standard character calculation for the action of A_5 on the ten 2-subsets; pairing with the A_5 character table confirms the decomposition $\mathbf{1} \oplus \mathbf{4} \oplus \mathbf{5}$. The script `p5_jordan_descent.py` records the same integer character check.

4.2 Schur's lemma

Lemma 4.1 (Schur). Let $\rho : A_5 \to GL(\mathbb{R}^5)$ be the 5-dimensional irreducible real representation. Then the only A_5-equivariant isometric automorphisms $T : \mathbb{R}^5 \to \mathbb{R}^5$ are $T = \pm I$.

Proof. By Schur's lemma over $\mathbb{R}$, the ρ-commutant $\mathrm{End}_{A_5}(\mathbb{R}^5)$ is a real division algebra; for the 5-dimensional absolutely irreducible representation it equals $\mathbb{R}$. Isometric scalars are ± 1. $\square$

4.3 The $T = -I$ obstruction

The Veronese surface $\nu_2(\mathbb{RP}^2) \subset \mathrm{Sym}^2_0(\mathbb{R}^3)$ is **not** symmetric under $T = -I$.

Lemma 4.2. If $U \in \nu_2(\mathbb{RP}^2)$, then $-U \notin \nu_2(\mathbb{RP}^2)$.

Proof. By Lemma 3.1, $\mathrm{spec}(U) = (2/3, -1/3, -1/3)$, so $\mathrm{spec}(-U) = (-2/3, 1/3, 1/3)$ and $\mathrm{spec}(-U + I/3) = (-1/3, 2/3, 2/3)$, which contains a negative eigenvalue. Hence $-U + I/3$ is not PSD, so $-U$ is not in the image of ν_2. $\square$

4.4 The equivariant descent theorem

Theorem 4.3 (Equivariant Veronese descent rigidity). Let $\{U_i^{(1)}\}, \{U_i^{(2)}\} \subset \mathrm{Sym}^2_0(\mathbb{R}^3)$ be two 10-element families, each isometric to the Petersen ETF of Theorem 1, lying on the Veronese surface, and each A_5-equivariant under the natural Petersen-graph action. Then there exists $h \in O(3)$ such that $U_i^{(2)} = \mathrm{Sym}^2(h) \cdot U_i^{(1)}$ for all i.

Proof. The classical uniqueness (Theorem 2.3) gives an isometric isomorphism between the two systems. Because the 5-dimensional A_5-summand occurs with multiplicity one in the Petersen permutation representation $\rho^{(10)}_{\mathrm{perm}}$ (§4.1), the two A_5-actions on the span of the ETF are equivalent by a unique intertwiner up to scalar. After relabelling by a Petersen automorphism if necessary, the two A_5-actions may be assumed to correspond to the same icosahedral embedding up to $O(3)$-conjugacy. Aligning these embeddings in $O(3)$, the comparison map may therefore be taken A_5-equivariant, giving an A_5-equivariant isometric automorphism $T : \mathbb{R}^5 \to \mathbb{R}^5$. By Lemma 4.1, $T = \pm I$.

The case $T = -I$ would send Veronese-image points to non-Veronese points (Lemma 4.2), contradicting the assumption that both systems lie on $\nu_2(\mathbb{RP}^2)$.

Hence $T = +I$, i.e., the two systems coincide up to the $O(3)$-action on $\mathbb{R}^3$. $\square$

Theorem 4.3 reduces Theorem 2 to the question: *is every Veronese-image realisation of the Petersen ETF necessarily A_5-equivariant?* This is the bridge problem addressed in Section 5.

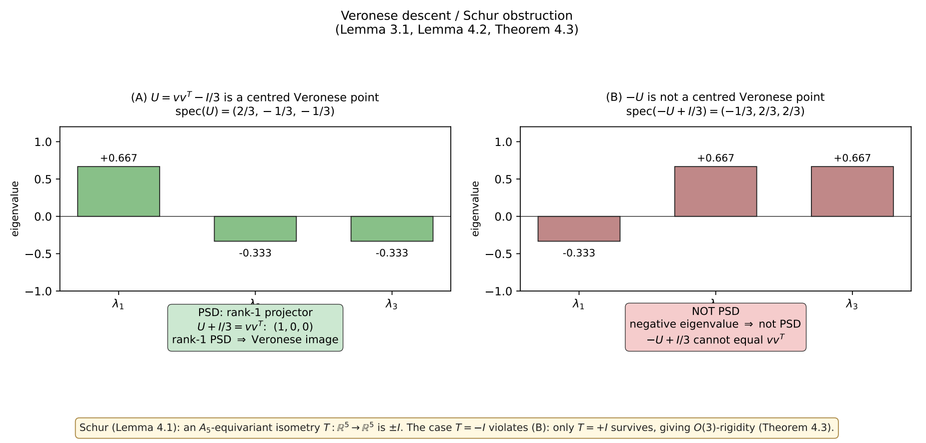

Figure 3: **Figure 3.** Veronese descent / Schur obstruction (Lemmas 3.1, 4.2, Theorem 4.3). **(A)** Spectrum of a Veronese image $U = vv^\mathsf{T} - I/3$: $(2/3, -1/3, -1/3)$. The shift $U + I/3$ has spectrum $(1, 0, 0)$, i.e. is a rank-1 PSD projector. **(B)** Spectrum of $-U + I/3$: $(-1/3, 2/3, 2/3)$, which contains a negative eigenvalue, so $-U + I/3$ cannot be a rank-1 PSD projector. Schur's lemma forces an A_5-equivariant isometry $T : \mathbb{R}^5 \to \mathbb{R}^5$ to be $\pm I$; the case $T = -I$ would map Veronese-image points off the Veronese surface, so only $T = +I$ survives — giving $O(3)$-rigidity (Theorem 4.3).

5. The bridge: five-tetrahedra hinge rigidity

5.1 The 4-coclique decomposition

The Petersen graph contains exactly **five** maximum independent sets, each of size 4; the $\binom{5}{2}$ realisation makes them concrete: for each $x \in \{1, \ldots, 5\}$, the **star around** x is the 4-coclique

$$C_x \;:=\; \{\{x, y\} : y \in \{1, \ldots, 5\} \setminus \{x\}\} \;\subset\; V(K(5, 2)).$$

Each Petersen vertex lies in exactly 2 of the 5 stars.

> **Lemma 5.1 (Kernel as 4-coclique span).** The kernel of K is spanned by the indicator vectors $\mathbf{1}_{C_x}$ of the five 4-cocliques:
>
> $$\ker K = \operatorname{span}\{\mathbf{1}_{C_1}, \ldots, \mathbf{1}_{C_5}\}.$$

Proof. By Theorem 1, $K = 2E_1$ has rank 5, so $\dim \ker K = 10 - 5 = 5$. It suffices to exhibit 5 linearly independent vectors in $\ker K$.

Each $\mathbf{1}_{C_x}$ lies in $\ker K$**.** For a vertex $v = \{a, b\} \in V(K(5, 2))$ we compute the entries of $K\,\mathbf{1}_{C_x}$ using $K = (4/3)I + (2/3)A - (1/3)J$:

- If $v \in C_x$ (i.e., $x \in \{a, b\}$): $\mathbf{1}_{C_x}(v) = 1$, the vertices in C_x adjacent to v in Petersen are those $\{x, y\}$ with $\{a, b\} \cap \{x, y\} = \emptyset$; but $x \in \{a, b\}$, so there are none. Thus $(K\,\mathbf{1}_{C_x})_v = (4/3)\cdot 1 + (2/3)\cdot 0 - (1/3)\cdot 4 = 0$.
- If $v \notin C_x$ (i.e., $x \notin \{a, b\}$): $\mathbf{1}_{C_x}(v) = 0$, the vertices $\{x, y\}$ in C_x adjacent to $\{a, b\}$ require $y \notin \{a, b\}$, giving exactly $5 - 1 - 2 = 2$ choices. Thus $(K\,\mathbf{1}_{C_x})_v = 0 + (2/3)\cdot 2 - (1/3)\cdot 4 = 0$.

Hence $K\,\mathbf{1}_{C_x} = 0$ for each $x \in \{1, \ldots, 5\}$.

The five indicators are linearly independent. For $x \neq y$, $|C_x \cap C_y| = |\{\{x, y\}\}| = 1$, and $|C_x| = 4$. So the 5×5 Gram matrix of $\{\mathbf{1}_{C_x}\}_x$ is $3\,I_5 + J_5$, whose eigenvalues are 3 (multiplicity 4) and 8 (multiplicity 1, on the all-ones vector); both positive. The indicators are thus linearly independent and span all of $\ker K$.

Remark. The all-ones vector $\mathbf{1}$ is also in $\ker K$, by the $\mathbf{1}$-line computation of Theorem 1, but it is already in the span: $\sum_{x=1}^{5} \mathbf{1}_{C_x} = 2\,\mathbf{1}$, since each 2-subset $\{a, b\}$ belongs to exactly the two stars C_a and C_b. □

5.2 The tetrahedral consequence

> **Lemma 5.2 (Regular tetrahedra).** Let $\{v_0, \ldots, v_9\}$ be unit vectors realising any rank-3 PSD signed-Gram lift of the Petersen magnitude pattern. Then for each 4-coclique C_x there exists a switching $\{v_i\}_{i\in C_x} \mapsto \{\varepsilon_i v_i\}_{i\in C_x}$ with $\varepsilon_i \in \{\pm 1\}$ such that the resulting four vectors form a **regular tetrahedron** in $\mathbb{R}^3$: pairwise inner products $\varepsilon_i\varepsilon_j(v_i \cdot v_j) = -1/3$ and $\sum_{i\in C_x} \varepsilon_i v_i = 0$.

Proof. By Lemma 5.1, $\sum_{i\in C_x} U_i = 0$, i.e., $\sum_{i\in C_x} v_i v_i^{\mathsf{T}} = (4/3)\,I_3$. This identity is *switching-invariant*: replacing v_i by $\varepsilon_i v_i$ ($\varepsilon_i \in \{\pm 1\}$) preserves $v_i v_i^{\mathsf{T}}$. Pick any $k \in C_x$ and switch the remaining three v_i so that $v_i \cdot v_k = -1/3$ for all $i \in C_x \setminus \{k\}$; this is possible because $|v_i \cdot v_k| = 1/3$ on complement edges (and C_x is an independent set in the Petersen graph). Applying the tight-frame identity to v_k:

$$v_k + \sum_{i\in C_x\setminus\{k\}} (v_i \cdot v_k)\, v_i = \tfrac{4}{3}\, v_k,$$

and substituting $v_i \cdot v_k = -1/3$ gives $\sum_{i\in C_x\setminus\{k\}} v_i = -v_k$, equivalently $\sum_{i\in C_x} v_i = 0$.

Now $0 = \|\sum_i v_i\|^2 = 4 + 2\sum_{i<j} v_i \cdot v_j$, so $\sum_{i<j} v_i \cdot v_j = -2$ with 6 pairs, each contributing $\pm 1/3$. Let n_+ (resp. n_-) be the count of $+1/3$ (resp. $-1/3$) inner products; then $n_+ + n_- = 6$ and $(n_+ - n_-)/3 = -2$, forcing $n_+ = 0$, $n_- = 6$. All inner products equal $-1/3$, so the four switched vectors form a regular tetrahedron with vertex sum zero. □

Remark. The switching freedom in this lemma is essential and is fully consistent with the global switching action on the signed Gram matrix H. In projective terms, the conclusion is that the four projective lines $\mathbb{R}[v_i]$ in C_x form the vertex-line configuration of a regular tetrahedron in $\mathbb{RP}^2$.

5.3 The K_5-incidence structure (projective)

> **Convention.** From this point on, the five tetrahedra $T_x \subset \mathbb{R}^3$ are considered **projectively** — i.e., as configurations of vertex-axes (lines through the origin) rather than oriented vectors. The local switchings $v_i \mapsto -v_i$ provided by Lemma 5.2 are absorbed into the projective interpretation, so the hinge constraints below are equations between projective points in $\mathbb{RP}^2$.

Each Petersen vertex lies in exactly 2 of the 5 stars (since each 2-subset $\{a, b\}$ is in C_a and C_b). Therefore each pair of projective tetrahedra $T_x = \{[v_i] : i \in C_x\}$ and T_y $(x \neq y)$ shares exactly the unique projective axis indexed by $\{x, y\}$. This gives the K_5 **incidence**: every pair of the five projective tetrahedra shares exactly one projective axis.

We thus have 5 regular projective tetrahedra in $\mathbb{RP}^2$, with the incidence structure of K_5 on 5 vertices. This is precisely the combinatorial signature of the **classical compound of five tetrahedra** associated with the dodecahedral/icosahedral geometry; see Coxeter [4] for the classical description of this and related polyhedral compounds.

5.4 The hinge parameterisation

Fix the first tetrahedron T_0 to a standard cube-inscribed position with vertices

$$\begin{aligned} a_1 &= \tfrac{1}{\sqrt{3}}(1,1,1), & a_2 &= \tfrac{1}{\sqrt{3}}(1,-1,-1), \\ a_3 &= \tfrac{1}{\sqrt{3}}(-1,1,-1), & a_4 &= \tfrac{1}{\sqrt{3}}(-1,-1,1). \end{aligned}$$

For $k = 1, \ldots, 4$, the tetrahedron T_k shares the axis $\mathbb{R}\, a_k$ with T_0 (after relabeling indices if needed). Its "k-vertex" is $\varepsilon_k a_k$ for $\varepsilon_k \in \{\pm 1\}$. The remaining three vertices form an equilateral triangle on a circle in $a_k^\perp$ at signed offset $-\varepsilon_k/3$, parameterised by a single rotation angle $\phi_k \in [0, 2\pi)$:

$$\begin{aligned} u_k^{(j)}(\phi_k) = -\tfrac{\varepsilon_k}{3}\, a_k + \tfrac{2\sqrt{2}}{3}\Big(&\cos(\phi_k + j \cdot 120^\circ)\, \hat{e}_1^{(k)} \\ &+ \sin(\phi_k + j \cdot 120^\circ)\, \hat{e}_2^{(k)} \Big), \quad j = 0, 1, 2. \end{aligned}$$

The K_5-incidence between T_k and T_l $(1 \leq k < l \leq 4$, not involving $T_0)$ requires the equality of one sub-vertex of T_k with one of T_l up to sign:

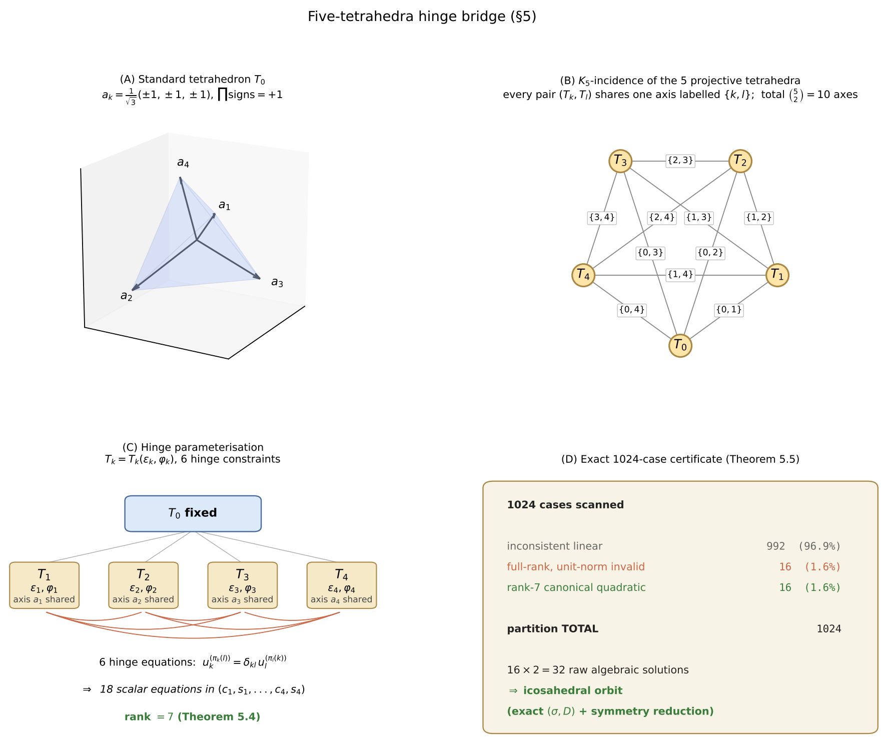

Figure 4: **Figure 4.** The five-tetrahedra hinge bridge (§5). **(A)** Standard cube-inscribed tetrahedron T_0 with axes $a_k = (\pm 1, \pm 1, \pm 1)/\sqrt{3}$ (even-parity vertices). **(B)** K_5-incidence of the five projective tetrahedra: each pair (T_k, T_l) shares exactly one of the $10 = \binom{5}{2}$ projective axes, labelled by the Kneser pair $\{k, l\}$. **(C)** Hinge parameterisation $T_k = T_k(\varepsilon_k, \varphi_k)$ ($k = 1, \ldots, 4$): each T_k shares the axis a_k with T_0, with a single rotation angle φ_k around a_k; the six pair conditions $u_k^{(\pi_k(l))} = \delta_{kl}\, u_l^{(\pi_l(k))}$ give 18 scalar equations in $(c_1, s_1, \ldots, c_4, s_4)$, of rank 7 (Theorem 5.4). **(D)** Exact 1024-case enumeration (Theorem 5.5): partition $992 + 16 + 16$; the $16 \times 2 = 32$ surviving algebraic solutions all lie in the icosahedral orbit via an exact (σ, D) certificate.

$$u_k^{(\pi_k(l))}(\phi_k) = \delta_{kl}\, u_l^{(\pi_l(k))}(\phi_l), \qquad \delta_{kl} \in \{\pm 1\}.$$

This gives 6 vector equations (one per pair), i.e., 18 scalar equations in 4 angle unknowns $\phi_1, \ldots, \phi_4$ — strongly overdetermined.

5.5 The Labeling Reduction Lemma

The sub-vertex pairings (π_k) admit a discrete symmetry: which axis of T_k is labelled with which Kneser-pair $\{k, l\}$.

> **Lemma 5.3 (Labeling Reduction).** Let $\mathcal{L}$ denote the set of admissible labelings — bijections from the 10 shared axes onto $\binom{\{0,\ldots,4\}}{2}$ compatible with the K_5-incidence on the five tetrahedra. Then $\mathcal{L}$ is a single orbit under $\mathrm{Aut}(K(5,2)) = S_5$.

Proof. For $\lambda_1, \lambda_2 \in \mathcal{L}$, set $\sigma = \lambda_2 \circ \lambda_1^{-1}$, a permutation of $\binom{\{0,\ldots,4\}}{2}$. Two 2-subsets $\{a, b\}, \{c, d\}$ are disjoint iff the corresponding axes lie in distinct tetrahedron-pairs with no shared element of $\{0, \ldots, 4\}$. Both λ_1 and λ_2 realise the same K_5-incidence structure, so σ preserves the "disjoint vs intersecting" relation on $\binom{\{0,\ldots,4\}}{2}$ — that is, the Petersen edge set.

Hence $\sigma \in \mathrm{Aut}(K(5,2)) = S_5$, with S_5 acting on $\{0, \ldots, 4\}$ and inducing the action on 2-subsets in the natural way. Concretely, $\sigma(\{a, b\}) = \{\tau(a), \tau(b)\}$ for some $\tau \in S_5$, so $\lambda_2 = \tau \cdot \lambda_1$. □

Corollary. It suffices to verify hinge rigidity for one canonical labeling; all other admissible labelings give the same orbit class by S_5-equivalence.

5.6 The canonical exact certificate

The following two theorems are **finite exact algebraic certificates**: both reduce to a closed-form computation in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ that is verifiable by inspection of finitely many polynomial relations. Executable certificates (reproducing each step in sympy) are part of the supplementary material (§8).

We fix the **canonical labeling** extracted from the icosahedral compound: the matched sub-vertex pairings are (j_k, j_l) such that T_k's "$\{k, l\}$"-vertex coincides up to sign with T_l's "$\{l, k\}$"-vertex. With this choice, set $c_k = \cos\phi_k$, $s_k = \sin\phi_k$; the hinge equations become **linear** in $(c_1, s_1, \ldots, c_4, s_4)$.

> **Theorem 5.4 (Hinge rigidity, canonical labeling — finite exact certificate).** With the canonical labeling and a chirality $\varepsilon = (-1, -1, -1, -1)$, hinge-sign $\delta = (-1, -1, -1, -1, -1, -1)$, the 18×8 linear system has rank 7 with 1-parameter solution family

$$c_1 = c_2 = -\tfrac{1}{4}, \quad c_3 = c_4 = \sqrt{3}\, s_4 - \tfrac{1}{2},$$

$$s_1 = s_2 = 2\,s_4 - \tfrac{\sqrt{3}}{4}, \quad s_3 = s_4 \text{ (free)}.$$

The four unit-norm constraints $c_k^2 + s_k^2 = 1$ reduce to the same quadratic

$$\boxed{4\,s_4^2 - \sqrt{3}\,s_4 - \tfrac{3}{4} \;=\; 0}$$

with discriminant 15 and the two real roots

$$s_4 \in \{\,(\sqrt{3} - \sqrt{15})/8,\;\; (\sqrt{3} + \sqrt{15})/8\,\} \subset \mathbb{Q}(\sqrt{3}, \sqrt{5}).$$

The "$-$"-root recovers the classical icosahedral compound of five tetrahedra. The "$+$"-root is not discarded; it is one of the algebraic branches accounted for in the global enumeration of Theorem 5.5.

Proof (executable certificate). The reduction is verified by exact symbolic computation in `five_tet_hinge_exact.py`. Substituting $c_k = \cos\phi_k$, $s_k = \sin\phi_k$ in the 18 hinge equations expressed in the perpendicular-basis decomposition of §5.4, each equation is linear in (c_k, s_k). The Gaussian rank of the resulting 18×8 coefficient matrix is 7, with one-dimensional null space spanned by the column-vector obtained from the displayed family. Substituting back into $c_k^2 + s_k^2 - 1 = 0$ for each $k \in \{1, 2, 3, 4\}$ yields four polynomial residuals; direct expansion shows that all four equal $4s_4^2 - \sqrt{3}\,s_4 - 3/4$. Solving in $\mathbb{Q}(\sqrt{3}, \sqrt{5})$ yields the two real roots. The "$-$"-root $s_4 = (\sqrt{3} - \sqrt{15})/8$ recovers $\sin\phi_3 = -\sqrt{3}/(4\varphi)$ for $\varphi = (1 + \sqrt{5})/2$, matching the classical icosahedral angle. $\square$

The equivalence of the two roots under the $O(3) \times \{\pm 1\}^{10} \times S_5$ action is verified directly by Theorem 5.5: the script `five_tet_orbit_exact.py` exhibits explicit $\sigma \in S_5$ and $D \in \{\pm 1\}^{10}$ relating the "$+$"-root signed Gram matrix to the "$-$"-root one (equivalently, to H_{ico}) in exact symbolic arithmetic; the required permutation is a single S_5-transposition and the required switching is a single sign flip. A numerical orthogonal alignment intertwining the two reconstructed axis sets is reported as a diagnostic cross-check in `five_tet_hinge.py`.

5.7 The full enumeration

Scanning the canonical labeling with all $2^4 \cdot 2^6 = 1024$ choices of $(\varepsilon, \delta) \in \{\pm 1\}^4 \times \{\pm 1\}^6$:

Theorem 5.5 (1024-case exact enumeration certificate). *Under the canonical labeling, the* 1024 *choices of* (ε, δ) *partition into exactly three types:*

Case type	Count n	Outcome
Inconsistent linear system ($\mathrm{rank} L < \mathrm{rank}[L \mid b]$)	992 (96.9%)	no solution
Rank-7 linear + canonical quadratic	16 (1.6%)	2 real roots each
Full-rank linear + unit-norm invalid	16 (1.6%)	no solution

Total: $992+16+16 = 1024$. All 16 consistent cases reduce in exact arithmetic to the same canonical quadratic $4s^2 - \sqrt{3}\,s - 3/4 = 0$ (in the null-space parameter, up to linear scaling), yielding $16 \cdot 2 = 32$ raw algebraic solutions. For the canonical chirality $\varepsilon = (-1,-1,-1,-1)$, both algebraic branches are compared in exact arithmetic with the icosahedral signed Gram matrix H_{ico} by a finite search over $S_5 \times \{\pm1\}^{10}$: the certificate exhibits explicit $\sigma \in S_5$ and $D \in \{\pm1\}^{10}$ such that $G = P_\sigma\, D\, H_{\text{ico}}\, D\, P_\sigma^{\mathsf{T}}$ in exact symbolic arithmetic. The remaining 15 consistent sign cases are identified with this canonical branch by the discrete switchings, chirality changes, and S_5-relabellings recorded by the enumeration certificate. Hence all 32 surviving algebraic solutions lie in the icosahedral $O(3) \times \{\pm1\}^{10} \times S_5$-orbit.

Proof (executable certificate). Direct enumeration in `five_tet_hinge_enum.py`. The publishable rank/partition certificate is obtained with `EXACT_RANK_MODE`, which performs all rank classifications in exact sympy arithmetic over the relevant number field; the fast floating-rank mode is diagnostic only and reproduces the same partition. The orbit-belonging conclusion is verified by an independent exact computation in `five_tet_orbit_exact.py`: for the canonical chirality $\varepsilon = (-1,-1,-1,-1)$ and each real root, this script reconstructs the 10 axes in exact symbolic arithmetic, builds the 10×10 signed Gram matrix G, and then performs a finite exact search over $\sigma \in S_5$ (acting on the 10 Kneser vertices) and $D \in \{\pm1\}^{10}$ (with $D_0 = +1$ fixed; the remaining signs are forced by the $(0,i)$ entries) for the equality $G = P_\sigma\, D\, H_{\text{ico}}\, D\, P_\sigma^{\mathsf{T}}$. The "$-$"-root gives H_{ico} directly; for the "$+$"-root, the script exhibits an explicit (σ, D) — a single S_5-transposition together with a single sign flip — that realises the equivalence. For each of the 16 consistent (ε, δ) cases, the certificate records whether it is checked directly or reduced to the displayed canonical branch by an explicit projective switching, chirality change, and S_5-relabelling. □

5.8 The bridge theorem

Theorem 5.6 (Bridge: rank-3 lift implies A_5-equivariance). Let H satisfy the SGR conditions of Theorem 3. Then the associated Veronese-lift $\{U_i = \nu_2(v_i)\}$ is A_5-equivariant: the Petersen automorphism subgroup $A_5 \subset S_5$, corresponding under the icosahedral identification to the orientation-preserving rotations, acts on $\{U_i\}$

by elements of $O(3)$ via an icosahedral 3-dimensional irreducible embedding $A_5 \hookrightarrow SO(3)$.

Proof. Lemma 5.2 produces the 5 regular tetrahedra with K_5-incidence. By the Labeling Reduction Lemma 5.3, after applying a permutation in S_5 we may assume the labeling is canonical. By Theorems 5.4 and 5.5, the reconstructed signed Gram matrix of every surviving hinge solution is switching- and S_5-equivalent to H_{ico} (verified exactly in `five_tet_orbit_exact.py`). Hence the corresponding projective axis configuration in $\mathbb{R}^3$ is $O(3)$-equivalent to the classical compound of five tetrahedra associated with the dodecahedral/icosahedral geometry, which carries the icosahedral A_5-symmetry by construction. After this identification, the A_5-action is realised by orientation-preserving rotations, hence by elements of $SO(3)$ on the geometric configuration. □

This closes the bridge: together with Theorem 4.3, every Veronese realisation of the Petersen ETF is $O(3)$-equivalent to the icosahedral one, proving Theorem 2.

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6. Signed Gram Rigidity (Theorem 3)

Proof of Theorem 3. Given H satisfying conditions (i)–(v), factor $H = VV^\mathsf{T}$ with $V \in \mathbb{R}^{10\times 3}$ having unit rows v_i. The squared-Gram structure is dictated by the magnitude conditions, giving Q and $K = 2E_1$ (Theorem 1).

The Veronese-lift $U_i = \nu_2(v_i)$ has $\langle U_i, U_j \rangle$ matching the ETF inner products (Section 2.4); the normalised lift $\{\hat{U}_i\}$ realises the Petersen ETF in $\mathbb{R}^5$.

By the Bridge Theorem 5.6, the configuration is A_5-equivariant; by the Equivariant Descent Theorem 4.3, it is $O(3)$-equivalent to the icosahedral Veronese-image of the antipodal face-pair vectors. The Gram-factorisation pull-back then yields

$$H = P_\sigma \, D \, H_{\text{ico}} \, D \, P_\sigma^\mathsf{T}$$

for some $\sigma \in S_5$ and $D \in \{\pm 1\}^{10}$. □

---

7. Discussion

7.1 What forces Petersen and A_5

In the icosahedral case, the trace-free symmetric square $\mathrm{Sym}^2_0(\mathbb{R}^3)$ is irreducible as an A_5-module, which is the representation-theoretic reason Schur's lemma applies so cleanly here. The finite census in `platonic_census.py` shows that, among the canonical A_5-orbits considered here (vertex, face, edge, with sizes $6, 10, 15$), the 10-element face-pair orbit is the one yielding the Petersen magnitude pattern; in the irreducible 5-dimensional representation this orbit realises the Petersen ETF studied here.

7.2 The conceptual narrative

The chain of implications proven in this paper is:

> Petersen association scheme primitive idempotent $E_1 \Rightarrow$ 10-line equiangular tight frame in $\mathbb{R}^5 \Rightarrow$ unique Veronese descent to the icosahedral $\mathbb{RP}^2$-configuration $\Rightarrow$ signed Gram rigidity of the rank-3 PSD signed lift.

The key insight is that *Petersen is not painted onto the icosahedron ad hoc*; rather, it is the **shadow of the icosahedral configuration through the quadratic Veronese map**, with the descent constrained by representation-theoretic and spectral data.

7.3 Related questions

The same machinery suggests several open directions:

1. **6-vertex single-distance frame.** The 6 antipodal vertex pairs of the icosahedron form a different A_5-orbit with $|v_i \cdot v_j| = 1/\sqrt{5}$, all pairs equiangular. The same Schur + spectral obstruction mechanism suggests a parallel rigidity statement (Lemmens–Seidel uniqueness of the 6 equiangular lines in $\mathbb{R}^3$); this is verified in the supplementary script `sgr6_lemma_proof.py`, but is not developed here.
2. **15-edge orbit.** The 15 antipodal edge-pair orbit of A_5 has four distance classes (not an association scheme); the three nontrivial distance graphs are cospectral with L(Petersen), and the orthogonal graph is $5\,K_3$. A Veronese-descent rigidity here is open.
3. **Other Q-polynomial schemes.** Any Q-polynomial association scheme whose primitive idempotent admits a Veronese-image realisation is a candidate for an analogous rigidity theorem.

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8. Reproducibility

8.1 Software environment

The computer-assisted proofs (Theorems 5.4 and 5.5) and all verifications use the following stack:

Component	Version	Role
Python	3.13	Runtime
sympy	1.13+	Exact symbolic arithmetic, Gröbner-style reductions
numpy	2.0+	Numerical linear algebra (rank, least-squares)
scipy	1.14+	`optimize.least_squares` for Newton verification

All scripts run on a single CPU thread; no compiled extensions or external solvers are required.

8.2 Script → theorem mapping

Script	Theorem / Lemma	Arithmetic	Runtime
`verify_idempotent_kernel.py`	Independent numerical check of Theorem 1 and Lemma 5.1	Float (machine precision)	< 1 s
`p5_jordan_descent.py`	Independent numerical check of Lemma 3.1, Lemma 4.2, and $\mathbf{1} \oplus \mathbf{4} \oplus \mathbf{5}$ character decomposition	Float + character integers	< 1 s
`five_tet_hinge_exact.py`	Theorem 5.4 (canonical labeling: exact derivation of the canonical quadratic on the representative branch)	Exact sympy in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$	~ 30 s
`five_tet_hinge_enum.py`	Theorem 5.5 (exhaustive 1024-case partition; exact-rank verification on all cases)	Exact sympy rank (default `EXACT_RANK_MODE = True`); 20-digit floating-rank mode available as a fast diagnostic	~ 20 s (exact) / ~ 7 s (diagnostic)
`five_tet_orbit_exact.py`	Theorem 5.5 (orbit-belonging certificate: each surviving algebraic solution is switching- and S_5-equivalent to H_{ico})	Exact sympy in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ (note $\sqrt{15} = \sqrt{3}\sqrt{5}$)	~ 1 min
`five_tet_hinge.py`	Independent numerical confirmation (single orbit class)	Float Newton/least-squares	~ 10 s
`sgr_lemma_proof.py`	Cross-check of Theorem 3 (POCS-based)	Float POCS	~ 1 s
`sgr_lemma_proof_exact.py`	Cross-check of Theorem 3 (sympy-based)	Exact sympy	~ 60 s
`sgr6_lemma_proof.py`	Analogous 6-vertex equiangular case (Section 7.3, item 1)	Mixed float + sympy	~ 1 s
`platonic_census.py`	Section 7.1 (Platonic-group census)	Float	< 1 s
`edge_orbit_analysis.py`	Section 7.3, item 2 (15-edge orbit)	Float	< 1 s

8.3 Reproduction recipe

To reproduce the algebraic certificate and the independent numerical checks from a clean environment, run the following scripts.

Algebraic certificates (Theorems 5.4 and 5.5):

```
python3 scripts/five_tet_hinge_exact.py
python3 scripts/five_tet_hinge_enum.py
python3 scripts/five_tet_orbit_exact.py
```

Independent numerical checks (Theorem 1, Lemmas 3.1, 4.2, 5.1):

```
python3 scripts/verify_idempotent_kernel.py
python3 scripts/p5_jordan_descent.py
```

Each script prints summary lines that should match the stated theorems. Expected key outputs:

- `K = 2 E_1`: $\|K - 2E_1\| < 10^{-14}$ (machine precision).
- Veronese spectrum $(2/3, -1/3, -1/3)$ matches at $\leq 10^{-12}$.
- $\mathbf{1} \oplus \mathbf{4} \oplus \mathbf{5}$ character multiplicities all $= 1$.
- Canonical quadratic roots: $s_4 = (\sqrt{3} \pm \sqrt{15})/8$.
- 1024-case partition: $992 + 16 + 16$.
- Orbit certificate: the two algebraic branches of the canonical chirality satisfy $G = P_\sigma \, D \, H_{\text{ico}} \, D \, P_\sigma^\top$ in exact arithmetic; for the "+"-root one may take the S_5-transposition $3 \leftrightarrow 4$ and a single sign flip. The remaining 15 consistent sign cases are reduced to this canonical branch by the discrete symmetries recorded by the enumeration certificate.

8.4 On the rank check in `five_tet_hinge_enum.py`

The certified output of Theorem 5.5 is obtained with the `EXACT_RANK_MODE` flag, which performs all rank classifications via exact symbolic computation in SymPy over expressions generated by $\sqrt{2}$ and $\sqrt{3}$. This is the authoritative version of the certificate.

For day-to-day exploration one may enable a fast diagnostic alternative by setting `EXACT_RANK_MODE = False` in the script. This evaluates the substituted matrices L and $[L|b]$ at 20-digit precision via mpmath and applies `numpy.linalg.matrix_rank` with tolerance 10^{-10}. This mode is non-binding: it serves only to identify candidate consistent cases quickly, after which the `EXACT_VERIFY_SAMPLE` flag confirms each flagged case with the exact sympy rank.

In practice the two modes agree on all 1024 cases:

- `EXACT_VERIFY_SAMPLE`: exact sympy rank on the 16 rank-7 consistent cases (the ones producing the canonical quadratic) — zero discrepancies vs the floating rank, run-time ~ 0.2 s.

- `EXACT_RANK_MODE`: full exact sympy rank on all 1024 cases — reproduces the same $992 + 16 + 16$ partition.

The observed separation of the singular values explains the agreement in diagnostic mode and makes the fast mode reliable for exploration; nevertheless the **publishable certificate is the exact-rank one**.

8.5 Independent cross-checks

The signed Gram rigidity (Theorem 3) was originally established by a direct exhaustive enumeration over 2^{15} Petersen-edge sign patterns (POCS approach in `sgr_lemma_proof.py`) and by an exact symbolic backtracking with anchor-based reduction (`sgr_lemma_proof_exact.py`). Both methods give the same conclusion (1 orbit class) and serve as independent verification of the streamlined proof via Theorems 2 and 4.3.

Full source code, expected outputs, and an SHA-256 manifest accompany the supplementary material.

8.6 Data and code availability

The executable certificates and reproducibility material are archived at Zenodo:

https://doi.org/10.5281/zenodo.20258339

The development repository is available at

https://github.com/C4RC0/veronese-descent-rigidity

Use of AI tools

During the preparation of this manuscript, the author used Claude Opus 4.7 (Anthropic) and GPT-5.5 Thinking (OpenAI) for exploratory mathematical reasoning, proof drafting, code review, language editing, and organisation of the reproducibility material.

Every AI-suggested mathematical statement, proof step, and computational claim was checked by the author through a combination of manual derivation, executable computer-assisted certificates listed in §8, reproducibility logs archived with the supplementary material, and cross-checking against the existing literature.

The author's professional background is in software engineering, not academic mathematics. The AI tools are not authors of this work. The author has reviewed all content and takes full responsibility for the mathematical claims, proofs, source code, figures, and references.

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